

SAP AI — Cheat Sheet Pack — Index & Glossary

How to use this pack: each sheet is one printable page — a concept, a minimal code pattern with line-by-line comments, Key Points, and Interview Q&A. Read in order 01→10; each stage builds on the one before it, mirroring the folders in your SAP AI Templates project.

Reading Order

#	Sheet	Answers the question
01	HANA DB Connection	How do I connect Python to SAP HANA Cloud?
02	LLM Setup — Gen AI Hub	How do I call an LLM through SAP AI Core?
03	NL → SQL (HANA AI Query)	How do I let AI write and run safe SQL?
04	HANA Vector Store	How do I store embeddings for semantic search?
05	RAG Pipeline	How do I ground LLM answers in my own documents?
06	SAP Chatbot Architecture	How do I combine LLM + RAG + SQL + memory into one bot?
07	Orchestration — Core	What is the Orchestration Service, templating, LLM module, filtering, masking?
08	Orchestration — Advanced	Grounding, structured output, history, full pipeline, agent
09	CAP + HANA Project Architecture	How is a real SAP full-stack AI project structured (CDS, services)?
10	CAP + AI Core + UI5	How does the CAP backend call AI Core, and how does UI5 talk to CAP?

Glossary — SAP AI Core / BTP Terms

Term	Meaning
SAP AI Core	The runtime service on BTP that hosts/serves AI models and pipelines.
Deployment	A running, callable instance of a model or scenario (has a deployment ID / URL).
Scenario	A category of workload in AI Core (e.g. foundation-models, orchestration).
Resource group	An isolation boundary inside one AI Core instance — separates teams/projects/environments.
Gen AI Hub (proxy)	A harmonized proxy API to call foundation models (OpenAI, etc.) directly — one model per call.
Orchestration Service	A managed pipeline in front of the LLM: templating → masking → filtering → grounding → LLM → filtering → unmask.
AI Launchpad	The web UI for managing AI Core: deployments, configurations, playground.
Service key	Credentials (AICORE_CLIENT_ID/SECRET/AUTH_URL/BASE_URL) generated in BTP Cockpit to authenticate.
CAP	Cloud Application Programming model — SAP's framework for building services on CDS + Node.js/Java.
CDS	Core Data Services — SAP's data modeling + query language, used for both DB schema and service definitions.
HDI container	HANA Deployment Infrastructure container — isolated schema/artifacts owned by one CAP app.
CQL	CDS Query Language — CAP's ORM-style query syntax (SELECT.from, INSERT.into, ...).

Term	Meaning
XSUAA	BTP's OAuth2/JWT authorization service — secures CAP services and UI5 apps.
Destination service	BTP service that stores target-system URLs + auth so app code doesn't hardcode credentials.
MTA	Multi-Target Application — a deployment descriptor (<code>mta.yaml</code>) bundling several modules (srv, db, app) into one deployable unit.
HANA Vector Engine	HANA Cloud's native vector data type + <code>COSINE_SIMILARITY()</code> function for semantic search.
REAL_VECTOR	HANA column type that stores an embedding vector of a fixed dimension.

Two Ways to Call an LLM on SAP AI Core

	Gen AI Hub Proxy	Orchestration Service
What you get	One model, one API call	A configurable pipeline around the model
Masking / filtering / RAG Best for	You write it yourself in Python Simple, single-model calls	Built into the platform config Governed, multi-step, auditable pipelines
Used in sheets	02, 03, 04, 05, 06	07, 08